Topic Title: Volatility Spillovers and Contagion Across Equities of European Financial Institutions.

Keywords: Volatility Spillovers, Contagion, Systemic Risk, Risk Diversification, Financial Stability.

Advisor:
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Expected Outcome: A publication in an economics, finance, applied mathematics, or econophysics journal

Language of Instructions, Reports: English

Your Main Tasks:

- Collect stock data from major European financial institutions (e.g., large, listed banks, insurance companies, hedge funds in Europe).
- Apply a range of statistical and econometric models (e.g., based on the Diebold and Yilmaz's approach and other multivariate models of volatility spillovers) to identify and extract volatility spillover linkages across their stocks over different time periods.
- Analyze the structure and properties of spillover network and its dynamics.
- Draw implications for the analysis of systemic risk, financial stability, and risk management.

Required Skills:

- Good academic writing skills in English.
- Experience in coding and working with financial data using *both R and Python*. Knowing MATLAB is a plus but not prerequisite.
- Strong background in quantitative methods.
- Basic knowledge of (multivariate) time-series econometrics and applied network science (or willing to learn). Experience with extracting API data and familiarity with Bloomberg Terminal are advantageous.

Additional Expectations:

- (Important!) effective time management and adherence to deadlines, a responsible and professional attitude.
- Self-discipline and initiative in studying relevant literature and learning the new methods and concepts.

Main References:

- Billio et al. (2012). Econometric measures of connectedness and systemic risk in the finance and insurance sectors. Financial Economics 104 (3): 535-559
- Diebold, F.X. and Yilmaz, K. (2009). Measuring Financial Asset Return and Volatility Spillovers, With Application to Global Equity Markets. Economic Journal 119, 158-171
- Diebold, F.X. and Yilmaz, K. (2012). Better to Give than to Receive: Forecast-Based Measurement of Volatility Spillovers. International Journal of Forecasting 28(1), 57-66
- Diebold, F.X. and Yilmaz, K. (2015). Trans-Atlantic equity volatility connectedness: U.S. and European financial Institutions, 2004-2014. Journal of Financial Econometrics 14.
- Wang et al. (2021). Multilayer information spillover networks: measuring interconnectedness of financial institutions. Quantitative Finance 21.
- And other competing multivariate models of volatility spillovers (e.g., multivariate GARCH and VAR variants, etc.) done in the pertinent literature.